# INFINITELY MANY EXCEPTIONS TO A PRIME-SQUARE REPRESENTATION PROBLEM OF ERDŐS

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Abstract. We prove that infinitely many positive integers have no representation $n = ap^2 + b$ with $p$ prime, $a \geq 1$, and $0 \leq b < p$. More precisely, the number of such integers in $(X, 2X]$ is of order $X/\log X$. The lower bound combines a positive-proportion avoidance estimate for large primes with first- and second-moment estimates for a short average of the small-prime sieve. A separated-frequency argument permits the large primes to be added in finitely many blocks. The small-prime moments have quadratic loss in the cutoff exponent, and a boundary estimate makes the large-prime avoidance stable under the averaging shifts.

## 1. The representation problem

Erdős asked whether every sufficiently large integer can be written as

$$n = ap^2 + b, \qquad p \text{ prime}, \quad a \geq 1, \quad 0 \leq b < p; \tag{1.1}$$

see [1, p. 68, problem 3]. Call an integer *exceptional* if it has no representation (1.1), and put

$$E(X) = \#\{n \in \mathbb{Z} : X < n \leq 2X,\ n \text{ is exceptional}\}.$$

**Theorem 1.1.** *There are constants $c, C > 0$ and $X_0$ such that*

$$c\frac{X}{\log X} \leq E(X) \leq C\frac{X}{\log X} \qquad (X \geq X_0). \tag{1.2}$$

*In particular, infinitely many positive integers have no representation* (1.1).

Erdős recorded the upper bound in [1, p. 68]. We prove the matching lower bound and recover the upper bound from the same sieve estimate.

We first work with positive integer $X$ and write $I = (X, 2X] \cap \mathbb{Z}$. For a prime $p$, let

$$A_p(n) = \mathbf{1}_{\{n \bmod p^2 < p\}}, \qquad V(z) = \prod_{p \leq z}\left(1 - \frac{1}{p}\right), \qquad S_z(n) = \prod_{p \leq z}(1 - A_p(n)).$$

The remainder $n \bmod p^2$ is taken in $\{0, \dots, p^2 - 1\}$. Euclidean division shows that a prime $p$ gives a representation (1.1) precisely when $p^2 \leq n$ and $A_p(n) = 1$. The associated intervals are $[ap^2, ap^2 + p - 1] \cap \mathbb{Z}$ for $a \geq 1$.

All sums and products indexed by $p$ run over primes. We use $\mathrm{e}(t) = \exp(2\pi i t)$ and write $\|t\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The notation $o_\alpha(1)$ means that $\alpha$ and all other exponent parameters have been fixed before $X$ tends to infinity.

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The classical prime estimates needed below are

$$\sum_{p\le t}\frac{1}{p}=\log\log t+B+o(1),\qquad V(t)\sim\frac{e^{-\gamma}}{\log t},\tag{1.3}$$

$$\pi(t)\ll\frac{t}{\log t},\qquad \vartheta(t):=\sum_{p\le t}\log p\ll t.\tag{1.4}$$

Here $\gamma$ is Euler's constant. These are Mertens' estimates and Chebyshev's bounds [2, 3]. Partial summation also gives

$$\sum_{p\le t}\frac{\log p}{p}\ll\log t,\qquad \sum_{p\le t}\frac{(\log p)^2}{p}\ll(\log t)^2.\tag{1.5}$$

The proof uses a fixed dyadic cutoff $z=X^\alpha$. Section 2 gives a positive proportion of points avoiding the larger primes. Section 3 estimates a short average of $S_z$ with relative error $O(\alpha^2)$ in both moments. Section 4 combines the estimates and proves the theorem.

## 2. Avoiding the large primes

We begin with the averaging estimate that permits separated blocks of primes to be added successively.

**Lemma 2.1** (Separated prime blocks)**.** *Fix $0<a\le b\le 1/4$ and $b<c\le 1/2$. Let $\mathcal{P}$ be any set of primes in $[X^a,X^b]$, and set*

$$G(n)=\sum_{p\in\mathcal{P}}A_p(n),\qquad \lambda=\sum_{p\in\mathcal{P}}\frac{1}{p}.$$

*Let $\mathcal{Q}$ be any set of primes in $[X^c,2\sqrt{X}]$, and let*

$$U(n)=\mathbf{1}_I(n)\prod_{q\in\mathcal{Q}}(1-A_q(n)).$$

*There is $\delta>0$, depending only on the fixed exponents, such that*

$$\sum_{n\in\mathbb{Z}}G(n)U(n)=\lambda\sum_{n\in\mathbb{Z}}U(n)+O(X^{1-\delta}).\tag{2.1}$$

*The error is uniform in $\mathcal{P}$ and $\mathcal{Q}$.*

*Proof.* Choose $b<\eta<c$, put $H=\lfloor X^\eta\rfloor$, and fix $0<\epsilon<\eta-b$. Take an even, nonnegative, smooth function $\phi$, supported in $[-1,1]$ and positive near zero. For large $H$,

$$\nu_H(h)=\frac{\phi(h/H)}{\sum_{r\in\mathbb{Z}}\phi(r/H)}\quad(h\in\mathbb{Z})$$

is a probability measure supported on $[-H,H]$. For each fixed positive integer $J$, finite differencing gives

$$\left|\sum_h\nu_H(h)\mathrm{e}(h\theta)\right|\ll_J(1+H\|\theta\|_{\mathbb{R}/\mathbb{Z}})^{-J}.\tag{2.2}$$

Indeed, the normalizing denominator is $\asymp H$, and the sum of the absolute $J$th differences of $\phi(h/H)$ is $O_J(H^{1-J})$. Multiplication by $(1-\mathrm{e}(\theta))^J$ gives (2.2), together with the trivial bound 1.

The Fourier expansion modulo $p^2$ is

$$A_p(n)=\sum_{k\bmod p^2}\widehat{A}_p(k)\mathrm{e}\big(kn/p^2\big),\qquad \widehat{A}_p(k)=\frac{1}{p^2}\sum_{r=0}^{p-1}\mathrm{e}\big(-kr/p^2\big).$$

In particular, $\widehat{A}_p(0) = 1/p$ and $|\widehat{A}_p(k)| \le 1/p$. Define $G_H(n) = \sum_h \nu_H(h)G(n+h)$ and put $D = X^\epsilon/H$. Discard from $G_H - \lambda$ the frequencies with distance greater than $D$ from an integer. Their total absolute contribution is

$$\ll_J X^{-J\epsilon} \sum_{p \le X^b} p \ll_J X^{2b-J\epsilon},$$

which is as small a power of $X$ as required after fixing $J$.

Every remaining nonzero frequency has a representative $k/p^2$ with

$$0 < |k| \le p^2 D < p$$

for sufficiently large $X$. It is therefore in lowest terms. Different labels give different points of $\mathbb{R}/\mathbb{Z}$, and their circular separation is at least $X^{-4b} \ge X^{-1}$. There are at most $2p^2D$ retained labels for each prime, so the sum of squared coefficients is

$$\ll \sum_{p \in \mathcal{P}} \frac{p^2 D}{p^2} \ll X^b D \ll X^{b+\epsilon-\eta}. \tag{2.3}$$

For completeness, a separated family of frequencies $\{\theta_r\}$ with spacing at least $X^{-1}$ satisfies

$$\sum_{n \in J_X} \left| \sum_r u_r \mathrm{e}(n\theta_r) \right|^2 \ll X \sum_r |u_r|^2 \tag{2.4}$$

on any interval $J_X$ of length $O(X)$. To prove this, choose a nonnegative smooth majorant $w(n/X)$ for the interval after translation. Twice summing by parts gives

$$\left| \sum_n w(n/X)\mathrm{e}(n\theta) \right| \ll X(1 + X\|\theta\|_{\mathbb{R}/\mathbb{Z}})^{-2}.$$

Spacing implies that each row sum of these bounds is $O(X)$. Expanding the square and using $2|u_r u_s| \le |u_r|^2 + |u_s|^2$ proves (2.4). Combining (2.3) and (2.4), and including the discarded frequencies, yields some $\delta_1 > 0$ for which

$$\sum_{X-H \le n \le 2X+H} |G_H(n) - \lambda|^2 \ll X^{1-\delta_1}. \tag{2.5}$$

Enlarge and contract both the ambient interval and every forbidden residue interval by $H$ at its endpoints. This gives indicators $U_-$ and $U_+$ such that, for $|h| \le H$,

$$U_-(n-h) \le U(n) \le U_+(n-h).$$

The enlarged ambient interval lies in $[X - H, 2X + H]$. For a prime $q$, the boundary change meets at most $O(H(X/q^2 + 1))$ integers in that interval. Thus

$$\begin{aligned} \sum_n (U_+(n) - U_-(n)) &\ll H + H \sum_{X^c \le q \le 2\sqrt{X}} (X/q^2 + 1) \\ &\ll HX^{1-c}, \end{aligned}$$

where the reciprocal-square tail can be bounded by the sum over all integers and $c \le 1/2$. Also $\lambda = O_{a,b}(1)$ by (1.3).

Since $G \ge 0$, averaging the pointwise comparisons and changing variables gives

$$\sum_n U_-(n)G_H(n) \le \sum_n U(n)G(n) \le \sum_n U_+(n)G_H(n).$$

Cauchy–Schwarz and (2.5) bound either centered sum by $O(X^{1-\delta_1/2})$. The boundary error is $O(HX^{1-c})$, which also saves a positive power because $\eta < c$. This proves (2.1). □

Put $d = 1 - \log 2$. We have $1/4 < d < 1$, since $\exp(3/4) > 1 + 3/4 + (3/4)^2/2 > 2$.

**Proposition 2.2.** *There is an absolute constant $c_0 > 0$ such that, for every fixed integer $L \geq 2$, with $\alpha = 2^{-L}$,*

$$\#\{n \in I : A_p(n) = 0 \text{ for every } X^\alpha < p \leq 2\sqrt{X}\} \geq c_0 d^L X \tag{2.6}$$

*for all sufficiently large $X$, depending on $L$.*

*Proof.* Each prime contributes $X/p + O(p)$ points of $I$ to its forbidden residues. Consequently the number of points surviving all primes $X^{1/4} < p \leq 2\sqrt{X}$ is at least

$$X - X \sum_{X^{1/4} < p \leq 2\sqrt{X}} \frac{1}{p} - O\left(\sum_{p \leq 2\sqrt{X}} p\right) = (d - o(1))X.$$

Here (1.3) gives the reciprocal sum $\log 2 + o(1)$, and (1.4) gives $\sum_{p \leq 2\sqrt{X}} p \ll X/\log X$.

Set $K = L - 2$ and $c_j = 2^{-j}/4$ for $0 \leq j \leq K$. Fix $0 < \tau < 1/2$ and add, in the order $j = 0, \ldots, K-1$, the blocks

$$X^{c_{j+1}} < p \leq X^{(1-\tau)c_j}.$$

At stage $j$, the previously imposed primes are all greater than $X^{c_j}$, so Lemma 2.1 applies. The reciprocal mass of the new block tends to $\log(2(1-\tau)) < \log 2$. If $N_j$ points survive before the block is added, the union bound and (2.1) give

$$N_{j+1} \geq dN_j - o(X).$$

There are only $K$ stages, all fixed before $X \to \infty$. The final selected-prime count is therefore at least $(d^{K+1} - o(1))X$.

The omitted gaps are $X^{(1-\tau)c_j} < p \leq X^{c_j}$. Their total reciprocal mass tends to $K[-\log(1-\tau)]$. Removing all their forbidden points costs at most

$$\big(K[-\log(1-\tau)] + o(1)\big)X;$$

their combined $O(p)$ errors are $o(X)$ by (1.4). Choose $\tau$, still at fixed $K$, so that $K[-\log(1-\tau)] < d^{K+1}/4$. For large $X$ at least $d^{K+1}X/2$ points remain. This gives (2.6) with $c_0 = 1/(2d)$. The case $K = 0$ follows directly from the initial estimate. □

## 3. The averaged small-prime sieve

Fix a small positive $\alpha$ and define

$$z = X^\alpha, \qquad \eta = \alpha(1 - \alpha^2), \qquad H = \lfloor X^\eta \rfloor, \qquad B(n) = \frac{1}{H} \sum_{j=0}^{H-1} S_z(n + j). \tag{3.1}$$

The following proposition supplies the two estimates used in the final argument.

**Proposition 3.1.** *There are absolute constants $C_1, C_2 > 0$ such that, for all sufficiently small fixed $\alpha > 0$,*

$$\sum_{n \in I} B(n) \geq (1 - C_1\alpha^2 - o_\alpha(1))XV(z), \tag{3.2}$$

$$\sum_{n \in I} B(n)^2 \leq (1 + C_2\alpha^2 + o_\alpha(1))XV(z)^2. \tag{3.3}$$

3.1. **The first moment.** Partition the primes through $z$ into nonempty blocks

$$\mathcal{P}_j = \{p : z^{2^{-j-1}} < p \le z^{2^{-j}}\}, \qquad 0 \le j \le J,$$

ending when the lower endpoint is below 2. Then $J = O(1+\log\log z)$. By (1.3), the sums $\sum_{p\in\mathcal{P}_j} 1/p$ are bounded by an absolute constant $C_0$, uniformly in $z, j$. Moreover,

$$V_j := \prod_{p\in\mathcal{P}_j}(1-1/p) \ge v_0 > 0 \tag{3.4}$$

uniformly, since $-\log(1-1/p) \le 2/p$ for $p \ge 2$.

Choose an absolute integer $L_0$ so large that $C_0^k/k! \le 2^{-k}$ for $k \ge 2L_0$, and put

$$T = \lceil \log_2(1/\alpha)\rceil + L_0, \qquad k_j = 2T + 2j.$$

For $K_j(n) = \sum_{p\in\mathcal{P}_j} A_p(n)$, the Bonferroni polynomials

$$U_j(n) = \sum_{i=0}^{k_j}(-1)^i\binom{K_j(n)}{i}, \qquad L_j(n) = \sum_{i=0}^{k_j-1}(-1)^i\binom{K_j(n)}{i}$$

satisfy

$$L_j \le \mathbf{1}_{\{K_j=0\}} \le U_j, \qquad U_j \ge 0, \qquad U_j - L_j = \binom{K_j}{k_j}.$$

Telescoping the product, using the nonnegativity of the upper polynomials, gives

$$S_z(n) \ge \prod_j U_j(n) - \sum_j (U_j(n) - L_j(n)) \prod_{i\ne j} U_i(n). \tag{3.5}$$

Replace the variables $A_p(n)$ temporarily by independent Bernoulli variables of means $1/p$. For a single block,

$$\mathbb{E}U_j \ge V_j, \qquad \mathbb{E}(U_j - L_j) \le \frac{C_0^{k_j}}{k_j!} \le 2^{-k_j}.$$

Also $\mathbb{E}U_j - V_j \le \mathbb{E}(U_j - L_j)$, so $\mathbb{E}U_j \le V_j(1 + v_0^{-1}2^{-k_j})$. Independence between blocks and (3.4) show that the expectation of the right side of (3.5) is at least

$$V(z)\left(1 - O\left(\sum_j 2^{-k_j}\right)\right) \ge V(z)(1 - O(\alpha^2)).$$

To transfer this bound to integers, expand (3.5) into products of distinct $A_p$. Every resulting squarefree modulus $m$ satisfies

$$m \le z^{\sum_{j\ge0} k_j 2^{-j}} = z^{4T+4} \le X^{1/4}$$

when $\alpha$ is sufficiently small. Each coefficient has absolute value at most $J+2$. The Chinese remainder theorem gives exactly $m$ forbidden residue classes modulo $m^2$ for a product indexed by $m$, hence its sum over $I$ is $X/m + O(m)$. The total transfer error is bounded by

$$O\left((J+2)\sum_{m\le X^{1/4}} m\right) = O((J+2)X^{1/2}) = o(XV(z)).$$

Thus $\sum_{n\in I} S_z(n) \ge (1 - C_1\alpha^2 - o_\alpha(1))XV(z)$. Translating $I$ by an integer $0 \le j < H$ changes the sum of $S_z$, which lies between zero and one, by at most $2H$. Since $H = o(XV(z))$, averaging these translates proves (3.2).

3.2. **A uniform two-position bound.** For an integer $k$, let $a_p(k)$ be the cardinality of the intersection of the forbidden residue sets at $n$ and $n+k$ modulo $p^2$. Set

$$g_k(p) = \frac{2}{p} - \frac{a_p(k)}{p^2}, \qquad V_k = \prod_{p \le z}(1 - g_k(p)).$$

Thus $g_k(p)$ is the density of the union of these two residue sets, and $1/p \le g_k(p) \le 2/p$.

**Lemma 3.2.** *For sufficiently small fixed $\alpha > 0$, uniformly in $k$ and in integer intervals $J_X$ of length $X$,*

$$\sum_{n \in J_X} S_z(n) S_z(n+k) \le (1 + C\alpha^2) X V_k + O(X^{3/5}), \tag{3.6}$$

*where $C$ and the implied constant are absolute.*

*Proof.* If $g_k(p) = 1$ for some $p$, then $V_k = 0$ and the left side is zero. We may therefore suppose $0 < g_k(p) < 1$ for every $p \le z$, and write $g(p) = g_k(p)$. Extend $g$ multiplicatively to squarefree products of primes through $z$, and put

$$q(r) = \prod_{p \mid r} \frac{g(p)}{1 - g(p)}, \qquad R = \lfloor X^{1/10} \rfloor, \qquad G_R = \sum_{r \le R} q(r).$$

All moduli in this proof are squarefree and supported on primes through $z$; in particular $g(1) = q(1) = 1$.

Let $F_r(n)$ indicate that the union event occurs for every $p \mid r$. For real weights $\lambda_r$ supported on $r \le R$, with $\lambda_1 = 1$, the square $(\sum_r \lambda_r F_r(n))^2$ majorizes $S_z(n) S_z(n+k)$. Its main term after summation over $J_X$ is

$$X Q(\lambda), \qquad Q(\lambda) = \sum_{r,s \le R} \lambda_r \lambda_s g([r,s]).$$

Writing $y_t = \sum_{t \mid r} g(r) \lambda_r$, one has

$$Q(\lambda) = \sum_{t \le R} \frac{y_t^2}{q(t)}, \qquad \lambda_1 = \sum_{t \le R} \mu(t) y_t. \tag{3.7}$$

The first identity follows by expanding $1/g((r,s)) = \sum_{t \mid (r,s)} 1/q(t)$, and the second is Möbius inversion. Choose $y_t = \mu(t) q(t) / G_R$. Then $\lambda_1 = 1$, $Q(\lambda) = 1/G_R$, and inversion gives

$$\lambda_r = \frac{\mu(r)}{G_R} \prod_{p \mid r} (1 - g(p))^{-1} \sum_{\substack{s \le R/r \\ (s,r) = 1}} q(s). \tag{3.8}$$

In particular $|\lambda_r| \le 1$. Indeed, the numerator without its sign equals $\sum_{s \le R/r, (s,r)=1} \sum_{e \mid r} q(se)$; the products $se$ are distinct and all at most $R$, so this is a subsum of $G_R$.

For $m = [r,s]$, the union conditions occupy $\prod_{p \mid m}(2p - a_p(k)) \le m^2$ residue classes modulo $m^2$. The Chinese remainder theorem therefore gives

$$\sum_{n \in J_X} F_m(n) = X g(m) + O(m^2).$$

There are at most $R^2$ pairs $(r,s)$ and $m \le R^2$. Using $|\lambda_r| \le 1$, the total error is $O(R^6)$, and hence

$$\sum_{n \in J_X} S_z(n) S_z(n+k) \le X / G_R + O(R^6). \tag{3.9}$$

For independent Bernoulli variables $Z_p$ with means $g(p)$,

$$G_R V_k = \mathbb{P}\left(\sum_{p\le z} Z_p \log p \le \log R\right).$$

The mean of the sum is $O(\log z)$, and its variance is $O((\log z)^2)$ by (1.5) and $g(p)\le 2/p$. Its second moment is therefore $O((\log z)^2)$. Markov's inequality applied to its square gives $G_R V_k \ge 1 - C'\alpha^2$ for large $X$. After making $\alpha$ smaller if necessary, $1/G_R \le (1+C\alpha^2)V_k$. Substitution in (3.9), with $R^6 \le X^{3/5}$, proves the lemma. □

3.3. **Averaging the local factors.** For $|k|<H$, put $w_H(k) = (H-|k|)/H^2$, and put $w_H(k)=0$ otherwise. These are the probabilities of the difference of two independent uniform integers in $\{0,\dots,H-1\}$. In particular, $\sum_k w_H(k) = 1$ and $w_H(k)\le 1/H$.

**Lemma 3.3.** *With the parameters (3.1),*

$$\sum_k w_H(k)\frac{V_k}{V(z)^2} \le (1-\alpha^2)^{-2} + o_\alpha(1). \tag{3.10}$$

*Proof.* Write

$$R_p(k) = \frac{1-g_k(p)}{(1-1/p)^2} = 1 + \frac{a_p(k)-1}{(p-1)^2}, \qquad v_p(k) = (R_p(k)-1)_+.$$

The function $R_p$ has period $p^2$, is nonnegative, and is at most $p/(p-1)$. Counting the pairs of forbidden residues gives $\sum_{k \bmod p^2} a_p(k) = p^2$, so $R_p$ has mean one over a period. Also

$$0 \le v_p(k) \le \frac{a_p(k)}{(p-1)^2}, \qquad \frac{1}{p^2}\sum_{k \bmod p^2} v_p(k) \le \frac{1}{(p-1)^2}, \qquad 1+v_p(k) \le \frac{p}{p-1}. \tag{3.11}$$

Choose a sufficiently small absolute $c_1>0$ and put

$$Y = c_1 \log H, \qquad M = \prod_{p\le Y} p^2, \qquad L(k) = \prod_{p\le Y} R_p(k).$$

By (1.4), $M \le H^{1/4}$. The Chinese remainder theorem shows that $L$ has mean one modulo $M$. A uniform integer in $\{0,\dots,H-1\}$ has residue probabilities $1/M + O(1/H)$; convolving with another probability distribution preserves this error bound. Consequently

$$\sum_k w_H(k)L(k) = 1 + O(M/H). \tag{3.12}$$

Set $Y_0 = H^{1/4}$ and $Y_1 = H^{1-\alpha^2}$. For $Y < p \le Y_0$, the nonnegative function $Lv_p$ has period $Mp^2$ and, by the Chinese remainder theorem, period mean $O(p^{-2})$. Counting complete periods and at most two incomplete periods in $|k|<H$, and using $w_H \le 1/H$, gives

$$\sum_k w_H(k)L(k)v_p(k) \ll p^{-2} + M/H.$$

By (1.3) and (3.11),

$$\prod_{Y<p\le Y_0}(1+v_p(k)) \ll \frac{\log Y_0}{\log Y}.$$

Telescoping this product and summing the preceding estimate show that its insertion after $L$ increases the weighted mean by at most

$$O\left(\frac{\log Y_0}{\log Y}\left(\frac{1}{Y} + \frac{MY_0}{H}\right)\right) = o(1).$$

For $Y_0 < p \le Y_1$, the same period-counting argument without $L$ gives

$$\sum_k w_H(k) v_p(k) \ll p^{-2} + H^{-1}.$$

The product of $1 + v_p$ in this range is $O_\alpha(1)$. The preceding product, including $L$, is pointwise $O(\log Y_0)$ by (1.3). A second telescoping estimate therefore bounds the additional weighted mean by

$$O_\alpha\left(\log Y_0 \left(\frac{1}{Y_0} + \frac{Y_1}{H}\right)\right) = o_\alpha(1).$$

Together with (3.12), this proves

$$\sum_k w_H(k) \prod_{p \le Y_1} R_p(k) \le 1 + o_\alpha(1).$$

Finally,

$$\prod_{Y_1 < p \le z} R_p(k) \le \prod_{Y_1 < p \le z} \frac{p}{p-1} = (1 + o_\alpha(1)) \frac{\log z}{\log Y_1}.$$

Since $\log H \sim \eta \log X$,

$$\frac{\log z}{\log Y_1} \longrightarrow \frac{\alpha}{(1-\alpha^2)\eta} = (1-\alpha^2)^{-2}.$$

All factors are nonnegative, so the last pointwise estimate can be multiplied into the preceding weighted bound. This proves (3.10). □

To finish Proposition 3.1, expand the square in (3.1). For each pair of shifts $i, j$, apply Lemma 3.2 on $I + i$ with $k = j - i$. Averaging the pairs and then using Lemma 3.3 gives

$$\sum_{n \in I} B(n)^2 \le (1 + C\alpha^2) X V(z)^2 \big((1-\alpha^2)^{-2} + o_\alpha(1)\big) + O(X^{3/5}).$$

For small $\alpha$, $(1-\alpha^2)^{-2} = 1 + O(\alpha^2)$, and $X^{3/5} = o(XV(z)^2)$ by (1.3). This is (3.3); the first moment was proved above.

## 4. Stable centers and the exceptional set

**Lemma 4.1.** *Let $\alpha = 2^{-L}$ with fixed $L \ge 2$, and let $H$ be as in (3.1). There is a set $U^* \subset I$ with*

$$|U^*| \ge c_2 d^L X \tag{4.1}$$

*for an absolute $c_2 > 0$ and all sufficiently large $X$, such that every $n \in U^*$ and $0 \le j < H$ satisfy*

$$n + j \in I, \qquad A_p(n + j) = 0 \quad (X^\alpha < p \le 2\sqrt{X}). \tag{4.2}$$

*Proof.* Start with the set counted in Proposition 2.2. Delete its points within $H$ of either endpoint of $I$, and delete the points meeting any forbidden residue interval enlarged by $H$ at both ends. A prime $p$ contributes at most $O(H(X/p^2 + 1))$ additional deleted points. The total loss is

$$O\left(H + H \sum_{X^\alpha < p \le 2\sqrt{X}} (X/p^2 + 1)\right) \ll H X^{1-\alpha} = o_\alpha(X),$$

because $\eta = \alpha(1-\alpha^2) < \alpha$ and $\alpha \le 1/4$. The remaining set satisfies (4.2), and (4.1) holds with $c_2 = c_0/2$. □

*Proof of Theorem 1.1.* We first prove the lower bound. Let $C_1, C_2$ be the constants in Proposition 3.1. Since $d > 1/4$,

$$\frac{(2^{-L})^2}{d^L} = (4d)^{-L} \longrightarrow 0.$$

Choose $L$ sufficiently large, put $\alpha = 2^{-L}$, and set $\rho = c_2 d^L$. We can require $0 < \rho \le 1$ and $\max(C_1, C_2)\alpha^2 \le \rho/16$. Then fix this $L$ and take $X$ sufficiently large to absorb all $o_\alpha(1)$ errors into $\rho/16$. Writing $V = V(z)$, we obtain

$$\sum_{n\in I} B(n) \ge (1-\rho/8)XV, \qquad \sum_{n\in I} B(n)^2 \le (1+\rho/8)XV^2, \qquad |I \setminus U^*| \le (1-\rho)X.$$

Cauchy–Schwarz now gives

$$\begin{aligned}\sum_{n\in U^*} B(n) &\ge \sum_{n\in I} B(n) - \sqrt{|I\setminus U^*| \sum_{n\in I} B(n)^2} \\ &\ge XV\left(1 - \frac{\rho}{8} - \sqrt{(1-\rho)(1+\rho/8)}\right) \ge \frac{\rho}{8}XV.\end{aligned}$$

The last inequality follows from $(1-\rho)(1+\rho/8) \le (1-\rho/4)^2$ for $0 \le \rho \le 1$.

Whenever $n \in U^*$ and a term $S_z(n+j)$ in $B(n)$ equals one, the integer $m = n+j$ lies in $I$ and avoids every prime through $z$. It also avoids the primes $z < p \le 2\sqrt{X}$ by (4.2). A larger prime has $p^2 > 2X$, so it cannot occur in a representation of $m$. Thus $m$ is exceptional. For any fixed $m$, at most $H$ pairs $(n, j)$ satisfy $m = n + j$ with $0 \le j < H$. Therefore

$$E(X) \ge \frac{1}{H} \sum_{n\in U^*} \sum_{j=0}^{H-1} S_z(n+j) = \sum_{n\in U^*} B(n) \ge \frac{\rho}{8} XV(z) \gg \frac{X}{\log X}$$

by (1.3).

For the upper bound, keep one sufficiently small fixed $\alpha$. For large $X$, every prime $p \le z$ satisfies $p^2 < X$. Every exceptional integer in $I$ is consequently counted by $S_z$. In Lemma 3.2, take $k = 0$. Then $S_z(n)^2 = S_z(n)$ and $V_0 = V(z)$, giving

$$E(X) \le \sum_{n\in I} S_z(n) \le (1 + C\alpha^2)XV(z) + O(X^{3/5}) \ll \frac{X}{\log X}.$$

This proves (1.2) for integer $X$. For real $X$, replacing $X$ by $\lfloor X \rfloor$ changes the integer interval $(X, 2X]$ by at most two points and changes $X/\log X$ by a factor tending to one. The same two bounds follow after adjusting the constants. □

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